

BMSE2003 Software Testing – Assignment

1. Objectives

This assignment provides students with the opportunity to learn and apply the following skills;

- To develop test plan, test cases, and test report.
- To use testing tools in executing a software testing process.

2. Assessment

The learning outcome assessed is:

CLO2: Adopt software test documentation standard in the development of test plan, test cases, and test report. (A3, PLO4)

CLO3: Utilise testing tools in executing a software testing process for the selected test object. (C3, PLO6)

Contribution of marks to coursework:

- The assignment deliverables* contributes **60%** (under CLO2 and CLO3) to the coursework component as follows:

Parts	Marks
❖ Create test plan, test cases, and test report	30
❖ Automation tools to design and execute test cases	20
❖ Apply standard ISO/IEC/IEEE 29119-3 on Test Artifacts	10
❖ Team collaboration, communication, presentation, and documentation	20
❖ Accessing digital skills in testing	20
Total	100

** Note: Tutor has the right to adjust the marks based on the % contribution by each team member.*

Refer to the Assignment Rubrics for details.

3. Group Formation

Students will be assigned to groups of **5 - 6** members per team from the same tutorial.

Every team member is expected to contribute and participate actively in the entire process of completing the assignment to develop complete documentation by creating a test plan, creating test cases, and performing appropriate testing techniques. Sharing of ideas and assistance in the completion of assignment among members is required.

4. Assignment Tasks

Students need to select **ONE (1)** published and currently functioning system, web, or mobile application as a sample to be tested and to create test documentations by applying software testing standard **ISO/IEC/IEEE 29119-3** in the test documentation. As part of the requirement, students are required to use any suitable automation testing tools in designing and executing the software testing process.

Students may not need to include ALL topics/sections in **ISO/IEC/IEEE 29119-3** for the report but the following topics **MUST** be included:

1. Test Plan.
2. Test Case (Appendix - FORM 1).
3. Test Execution Summary Report.

5. Practical Schedule

Week	Practical Task	Deliverable
1	Discussion on assignment question, assignment task, and assignment group formation	Week 1's task(s)
2	Discussion on the selected system, web or mobile application for assignment. List the individual modules and test items to be tested	Week 2's task(s)
3	Discussion on the implementation of ISO/IEC/IEEE 29119-3 standard for Test Plan, Test Case Specification, Test Execution Summary Report preparation	Week 3's task(s)
4	Discussion on the context of testing of the Test Plan	Week 4's task(s)
5	Discussion on the risk register of the Test Plan	Week 5's task(s)
6	Discussion on the test strategy of the Test Plan – test design technique(s)	Week 6's task(s)
7	Discussion on the test strategy of the Test Plan – test entry and exit criteria	Week 7's task(s)
8	Discussion on the test strategy of the Test Plan – test environment, testing tool, test data	Week 8's task(s)
9	Discussion on the testing activities and estimates of the Test Plan	Week 9's task(s)
10	Discussion on the staffing and schedule of the Test Plan	Week 10's task(s)
11	Discussion on Test Case Specification	Week 11's task(s)
12	Discussion on Test Execution Summary Report	Week 12's task(s)
13	Review on Test Plan, Test Case Specification, Test Execution Summary Report. Assignment Submission	Week 13's task(s)
14	Feedback on the assignment	Week 14's task(s)

6. The Final Assignment Report Format

The final assignment report for all Parts should contain the following items:

- Cover sheet (Appendix - FORM 2)
- Plagiarism statement with student signatures (Appendix - FORM 3)
- Assignment Rubrics (Group and Individual)
- Table of Contents
- Body of answers
- Appendices (if any)

The report must be type-written using **MS-Word**. You are recommended to format your report according to the following specification:

Media	Softcopy (.pdf) - well written and properly formatted report to be submitted via uploading softcopy to Google Classroom which include all work/deliverables for Assignment report.
Font Size	A body text of font <i>size 12</i> is required while for headings and subheadings a larger font size must be used.
Font Style	Use <i>Times New Roman</i> for body text. Main headings and sub-headings should be clearly stated using suitable font styles (e.g. Arial).
Line Spacing	Typed material should be <i>1.5-line spaced</i> .
Alignment	Use <i>Justify</i> for alignment.
Headers and Footers	Appropriate footers and headers should be used to enhance clarity and presentation.
Page Numbering	Ensure that all pages (except cover page) are numbered.
Paper Size	Use <i>A4</i> paper (29.7cm x 21cm).

Table 1: Written Report Format

7. Late Submission

All assignments should be submitted by **Week 13, Monday, 11.59 pm.**

Late submission of the assignment will be handled according to the Guideline for Late Submission of Coursework available at **TAR UMT's Intranet Examinations and Credit Accumulation Undergraduate Guideline.**

PART B : LATE SUBMISSION OF COURSEWORK INFORMATION

Please check/tick one of the information below :

☐	Late submission of 1 - 3 days after deadline of submission: minus 10 marks
☐	Late submission of 4 - 7 days after deadline of submission: minus 20 marks
☐	Late submission of > 7 days after deadline of submission: 0 mark

In certain circumstances, a student may be allowed to submit the assignment late with valid reason. S/he must inform the respective tutor **at least one week before** the assignment is due. The tutor will evaluate whether the circumstance warrants submitting the assignment late, but **no guarantee** that the students will not be penalized.

As a general rule, no extension of time will be granted. The assignment question and its due dates are normally disclosed in advance to students in order that they will be able to manage their time according to different course study progress and complete this assignment on time.

Note:

- a) For EMC (Extenuating Mitigating Circumstance) cases, students need to apply the Leave of Application from the FOCS office as a record where proof of MCs, Death Certificate, SPM Examination, MUET Examination, etc. is given.
- b) For Late Submission cases, students need to fill up the "Late Submission of Coursework" form which can be downloaded from the Intranet as shown in (Appendix - FORM 4).
- c) For free-rider cases, students need to fill up the "Group Contract Form" for free-rider case which can be downloaded from the Intranet as shown in (Appendix - FORM 5).

8. Academic Integrity and Plagiarism

Students are required to redo the entire assignment if the contents of any part of the assignment does not comply with the requirements of the assignment as requested by the respective lecturer / tutor.

Before submitting your assignment, please make sure that you have complied with **TAR UMT Plagiarism Policy**. Any cheating, attempt to cheat, plagiarism, collusion and any other attempts to gain an unfair advantage in assessment will cause the students concerned to be penalized.

IMPORTANT: Students found to be dishonest are liable to disciplinary action.

Test Case Template	
Test Case #:	Test Case Name:
System:	Subsystem:
Design By:	Design Date:
Executed By:	Execution Date:
Short Description:	

Pre-conditions:

Step	Action	Expected System Response	Pass/Fail	Comments
1				
2				
3				
4				
5				

Post-conditions:



FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

Programme: _____ (Group: _____)

Assignment

BMSE2003 SOFTWARE TESTING

Name (Block Letters)	Registration No.	Signature	Marks

Tutor's Name: _____

Date of Submission: _____



FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

Plagiarism Statement and Guideline for Late Submission of Coursework

Read, complete, and sign this statement to be submitted with the written report.

We confirm that the submitted work is all our own work and are in our own words.

Name (Block Letters)	Registration No.	Signature & Date	Picture

LATE SUBMISSION OF COURSEWORK

PART A: STUDENT INFORMATION

Name:	ID:
Faculty/Centre/Branch:	
Programme:	Year/Semester of Study:
Course Code & Course Title:	
Assignment Title:	

Please check/tick one of the information below:

	Late submission of 1 - 3 days after deadline of submission: minus 10 marks
	Late submission of 4 - 7 days after deadline of submission: minus 20 marks
	Late submission of > 7 days after deadline of submission: 0 mark

By signing below, I acknowledge that my coursework marks shall be subjected to the above penalty.

Student's Signature

Date _____

I acknowledge that I received the above student's coursework.

Lecturer / Tutor Name:

Signature:

Date _____

Received by Faculty/Centre/Branch

(Signature)

Date _____

Name :

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY
Group Contract Form

Faculty:
Programme:
Year of Study:
Semester:
Course Title:
Course Code:
Group Assignment/ Project Title:
Name of Tutor:
Tutorial Group:

Student Name and ID	Role(s)/Task(s) Assigned	Signature and Date